Latest Trends in Cloud and DevOps



Latest Trends in DevOps Practices



Engage and Think



Imagine you are part of a team navigating today’s fast-moving work environment—one where delivering quickly, communicating clearly, and adapting confidently is critical. Your team wants to improve how it works together, whether through better planning, clearer handoffs, or faster feedback.

If you could change one habit or process within your team to make your work more efficient and collaborative, what would you choose and why would it matter?

Learning Objectives

By the end of this lesson, you will be able to:

- Apply GitOps principles to streamline automated deployment workflows using version-controlled manifests
- Identify emerging DevOps trends (such as AIOps, Low-Code, and IaC) that align with organizational goals for improving software delivery efficiency
- Construct an infrastructure provisioning strategy using Infrastructure as Code (IaC) for managing scalable, cloud-native environments
- Develop a use case to demonstrate how AIOps reduces incident resolution time through real-time anomaly detection and root cause analysis
- Utilize low-code platforms to build and integrate business-specific applications with minimal manual coding



Latest Trends in DevOps Practices

What Is DevOps?

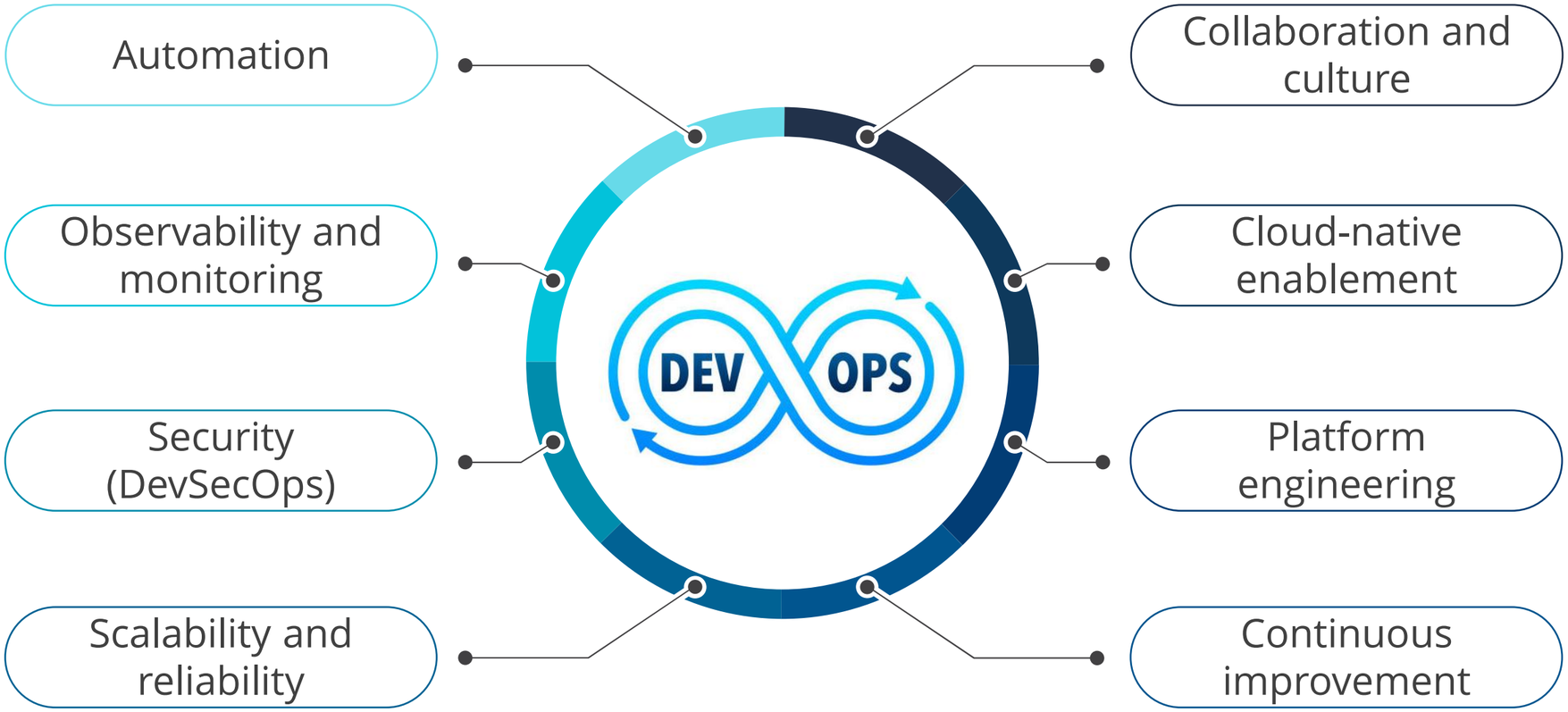
It is a modern software delivery approach that integrates development, operations, security, and automation to enable continuous, reliable, and scalable application delivery.



It emphasizes platform engineering, GitOps, infrastructure as code, AIOps, observability, DevSecOps, and cloud-native practices to achieve faster and more secure releases.

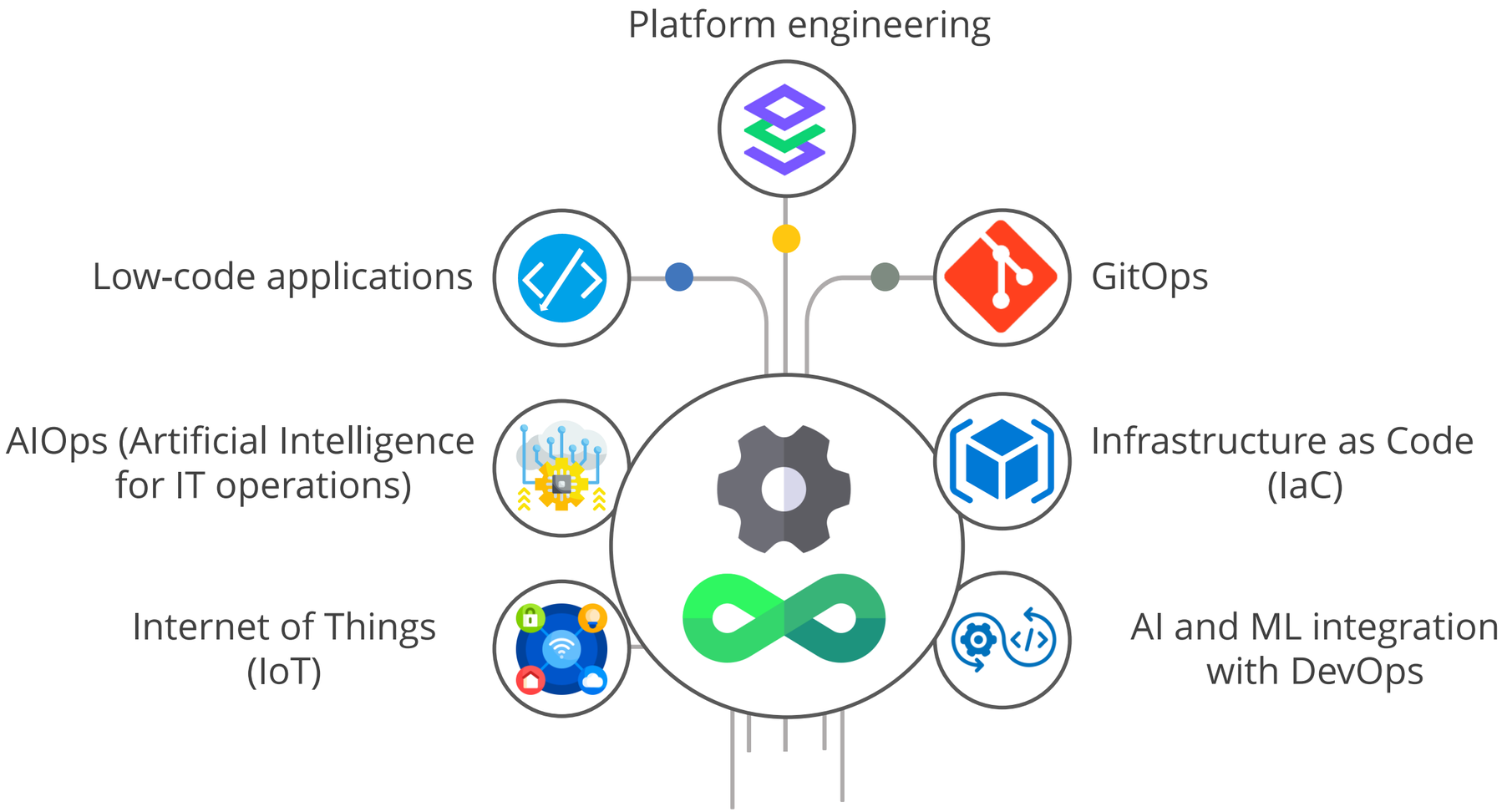
Key Focus Areas in DevOps

The core focus of DevOps is to deliver software faster, more securely, and efficiently by leveraging culture, automation, and continuous innovation. The key focus areas include:



What Are Latest Trends in DevOps Practices?

The latest trends in DevOps involve emerging practices and technologies that enhance collaboration, automation, and performance throughout the software delivery lifecycle. Key trends in DevOps include:



Quick Check



A global company aims to reduce release time, enhance service reliability, and scale efficiently during peak demand. Which DevOps focus areas would best support these goals?

A. Monitoring, low-code, manual approvals

B. Automation, CI/CD, platform engineering, scalability

C. Documentation, GitHub issues, weekly releases

D. Email alerts, patching, monoliths

Overview of Platform Engineering

What Is Platform Engineering?

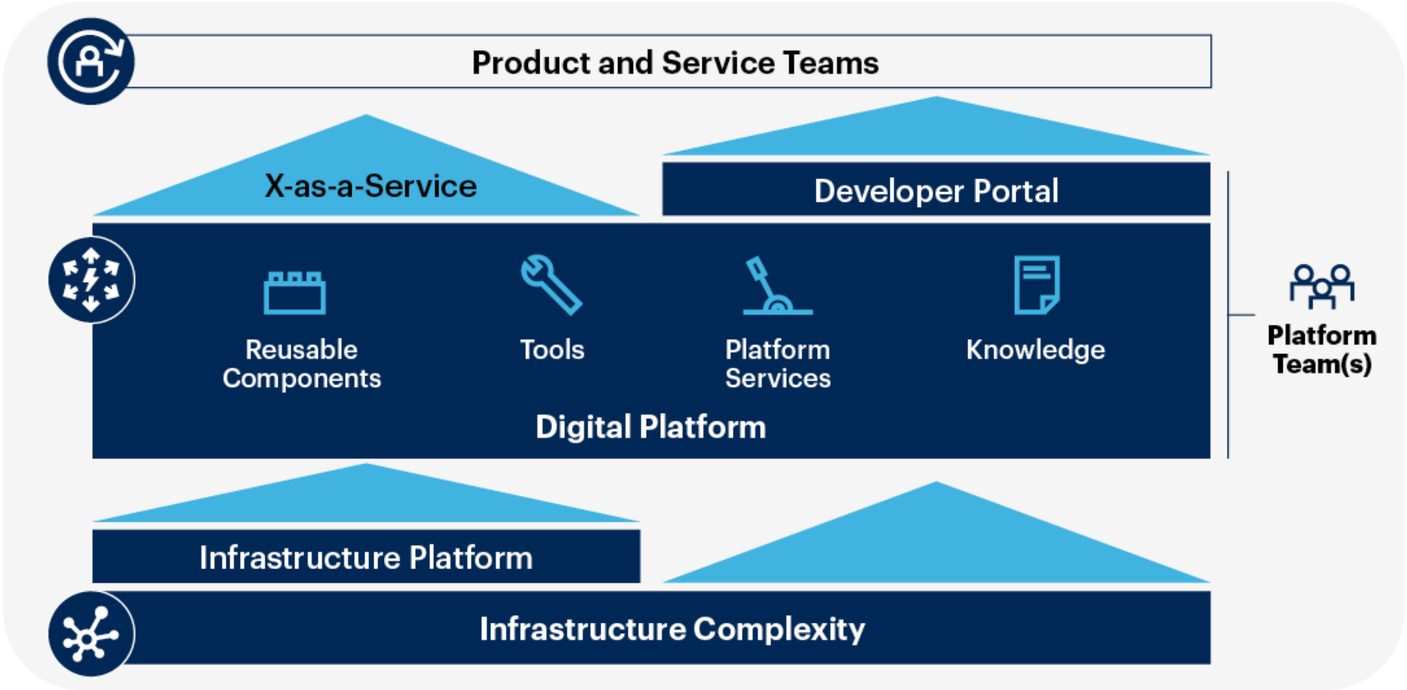
It is a practice derived from DevOps principles. It aims to enhance security, compliance, cost-efficiency, and the time to value for development teams.



To achieve these goals, platform engineering builds internal developer platforms that offer self-service, automation, and reusable components, reducing complexity and improving developer productivity.

Anatomy of a Platform Engineering Architecture

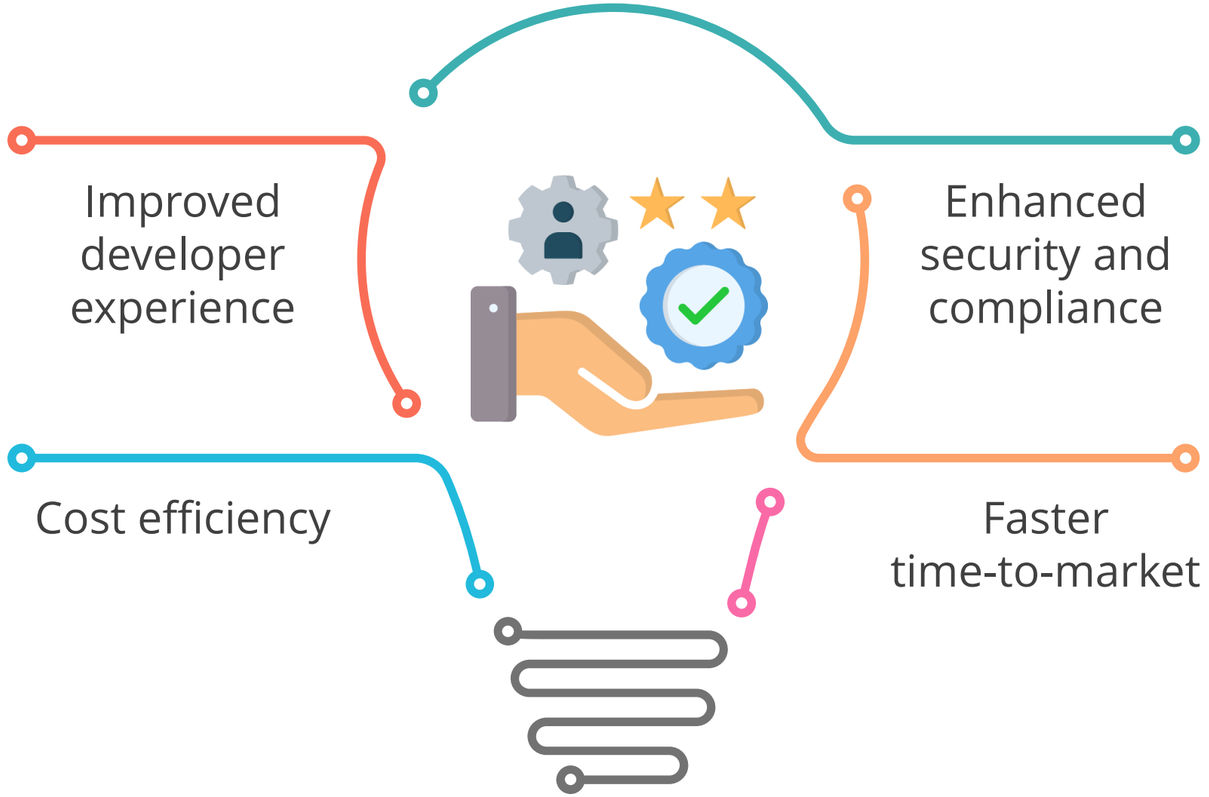
Platform teams enable product and service teams to innovate faster by abstracting infrastructure complexity and delivering reusable, self-service capabilities.



By bridging the gap between infrastructure and development needs, digital platforms enable scalable, efficient, and secure software delivery across the enterprise.

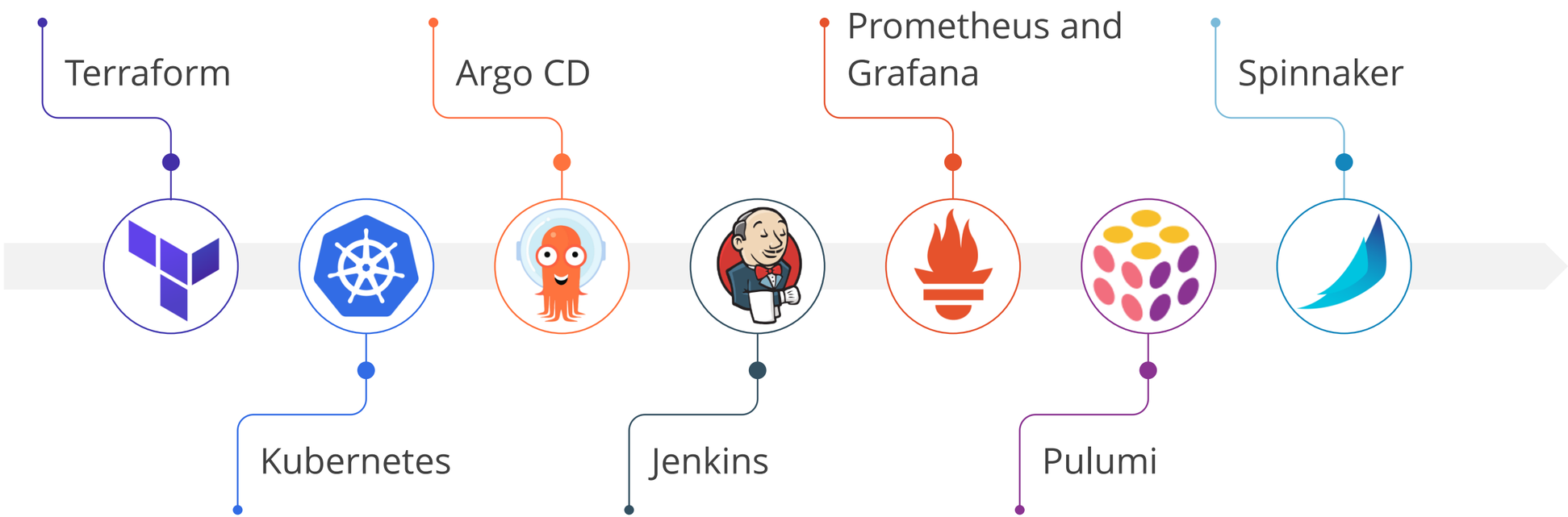
Benefits of Platform Engineering

Platform engineering offers several key benefits for modern software teams, including:



Platform Engineering Tools

The following tools form the backbone of efficient, scalable, and automated platform operations enabling teams to deliver faster, safer, and smarter:



Quick Check



A development team needs to deploy services quickly without managing infrastructure complexity. The platform team creates a developer portal that includes X-as-a-Service, tools, and reusable components. Which platform engineering layer is being used to abstract complexity?

A. Product and service layer

B. Monitoring stack

C. Infrastructure platform

D. Internal developer platform

Exploring GitOps

What Is GitOps?

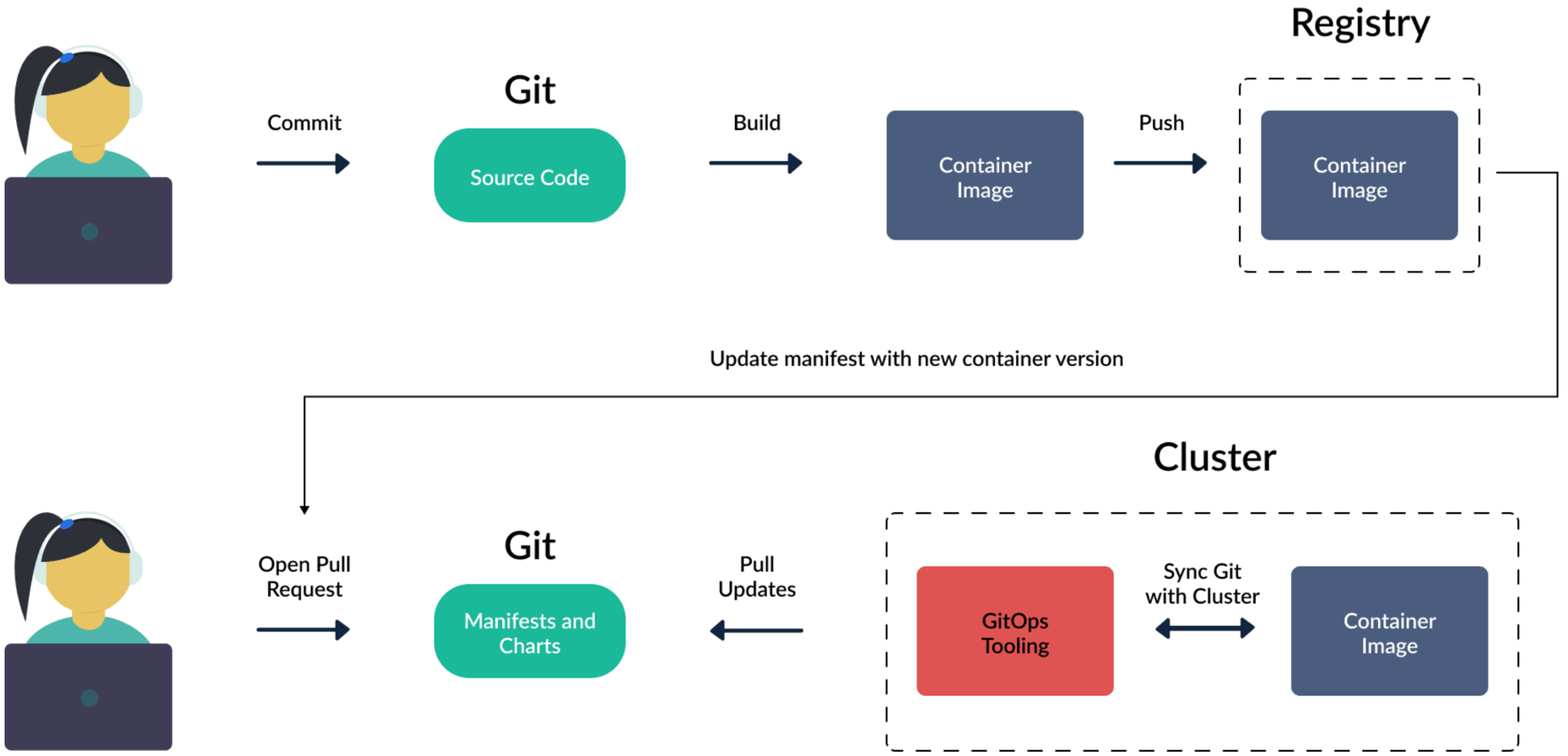
It is an operational framework that builds on DevOps practices like continuous integration and continuous delivery (CI/CD) and version control to automate infrastructure and manage software deployment.



It enhances collaboration between development and operations teams by using familiar Git workflows to manage and deploy infrastructure.

GitOps Workflow: Automating Deployment from Code to Cluster

The following model enables secure, auditable, and consistent application delivery by synchronizing the infrastructure state with version-controlled manifests.



GitOps vs DevOps

Understanding the key differences between GitOps and DevOps helps teams choose the right strategy to streamline software deployment and operations. Below is a comparison between them:

Category	GitOps	DevOps
Approach	Uses a declarative, version-controlled deployment strategy	Uses a flexible, tool-agnostic deployment strategy
Tools	Primarily relies on Git-based tools	Utilizes a variety of CI/CD pipeline tools
Focus	Prioritizes accuracy, consistency, and traceability	Prioritizes speed, automation, and frequent delivery
Correctness	Emphasizes precision, auditability, and compliance	Focuses less on precision and more on delivery speed
Flexibility	Enforces stricter workflows and policies	Supports more open and adaptable workflows

Benefits of GitOps

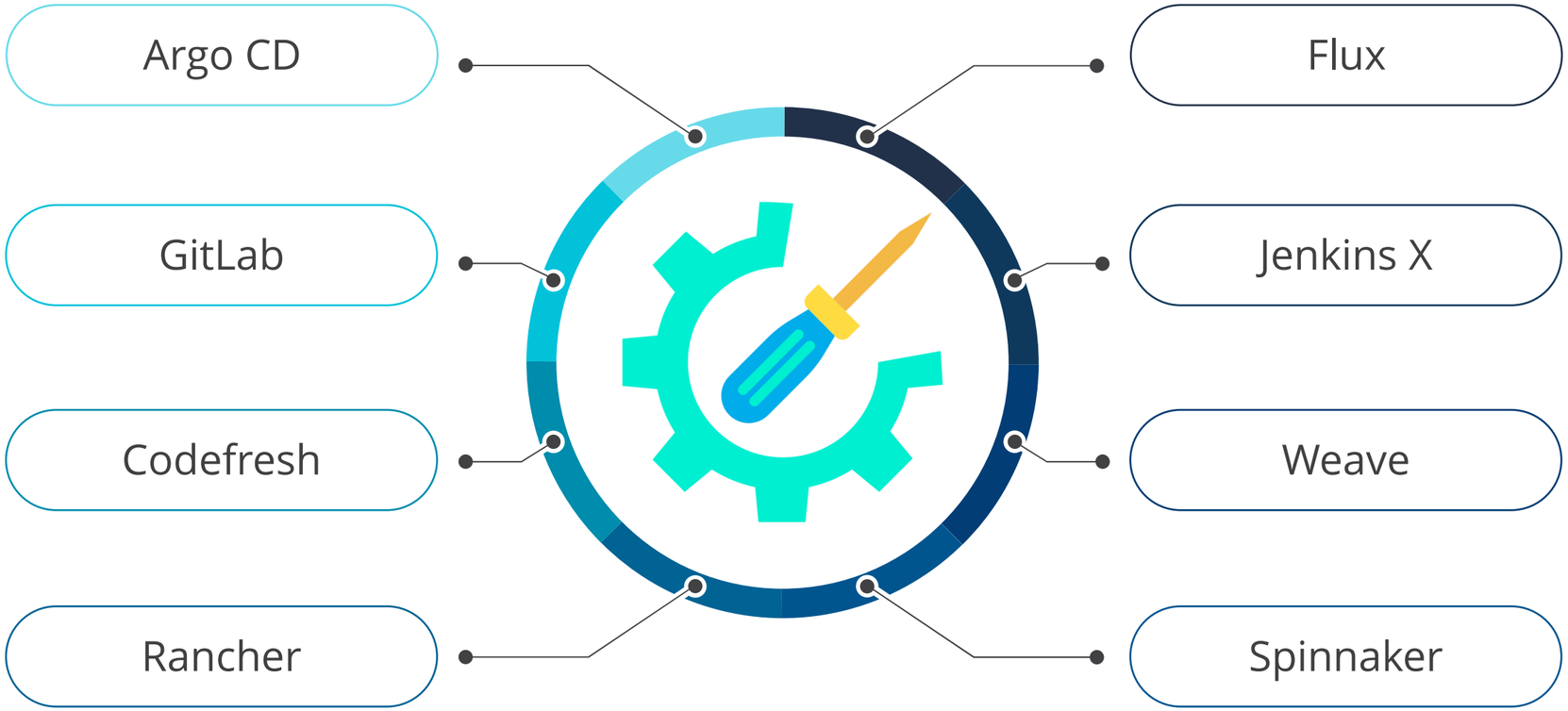
GitOps offers the following benefits:

- Consistency and reliability
- Auditability and traceability
- Improved collaboration
- Automation and efficiency
- Security and compliance

GitOps strengthens software delivery by improving collaboration, enhancing deployment reliability, and providing greater visibility and control across environments.

Tools Used in GitOps

Popular tools that power GitOps workflows across modern DevOps pipelines include:



Quick Check



A developer commits code, builds a container image, pushes it to the registry, and updates the deployment manifest in Git. According to GitOps principles, what should happen next to update the cluster automatically?

A. Manually apply changes using kubectl

B. Push the image directly into the cluster

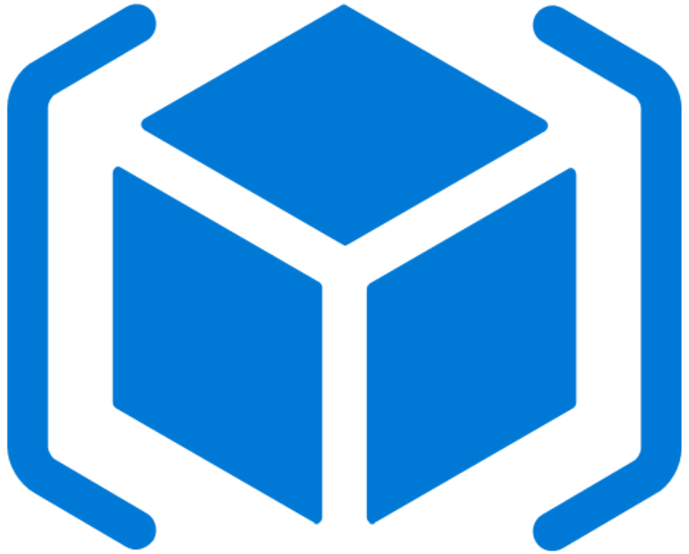
C. Sync the cluster using a GitOps tool

D. Rerun CI/CD unit tests before deployment

Understanding Infrastructure as Code (IaC)

What Is Infrastructure as Code (IaC)?

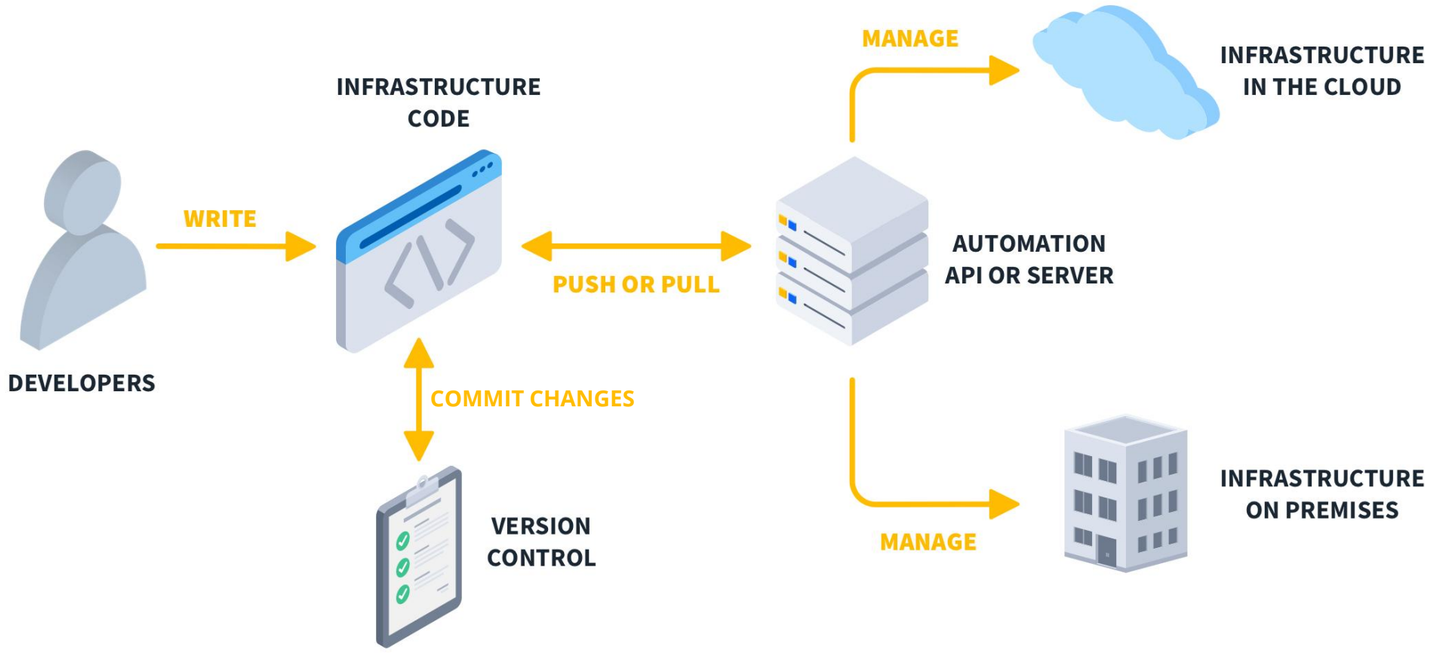
It is the practice of provisioning and managing infrastructure using machine-readable configuration files instead of manual processes.



It enables DevOps teams to collaborate using a unified set of practices and tools to deliver applications and infrastructure at scale.

Modern Infrastructure Management Using IaC

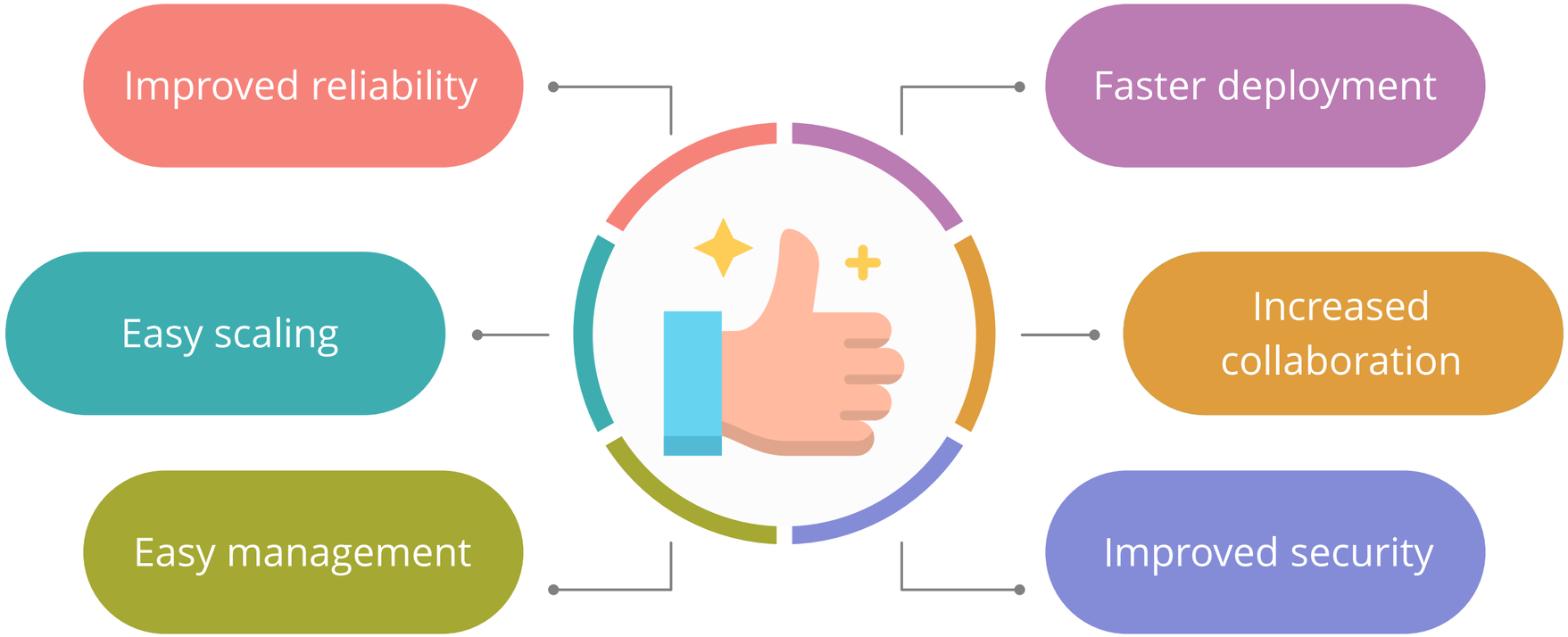
IaC enables developers to define, version, and automate infrastructure as code, helping bridge the gap between development and operations.



This approach allows infrastructure whether in the cloud or on-premises to be consistently deployed and managed through automated servers and version-controlled code.

Advantages of IaC

Infrastructure as Code (IaC) improves infrastructure delivery by offering the following key advantages:



Use Cases of IaC

Infrastructure as Code supports a wide range of use cases that streamline and automate infrastructure management tasks. Below are some notable examples:



From provisioning VMs to managing DNS records, IaC ensures repeatability, speed, and consistency throughout the infrastructure lifecycle.

Latest Trends in Infrastructure as Code

IaC continues to evolve with innovations that enhance scalability, security, and developer autonomy across modern cloud environments. Below are some of the notable trends:



Quick Check



A DevOps team needs to manage infrastructure consistently across both cloud and on-premises environments. They plan to use version-controlled files and automated APIs for deployment. Which practice best supports this approach?

A. Manual configuration using CLI scripts

B. Infrastructure as Code (IaC)

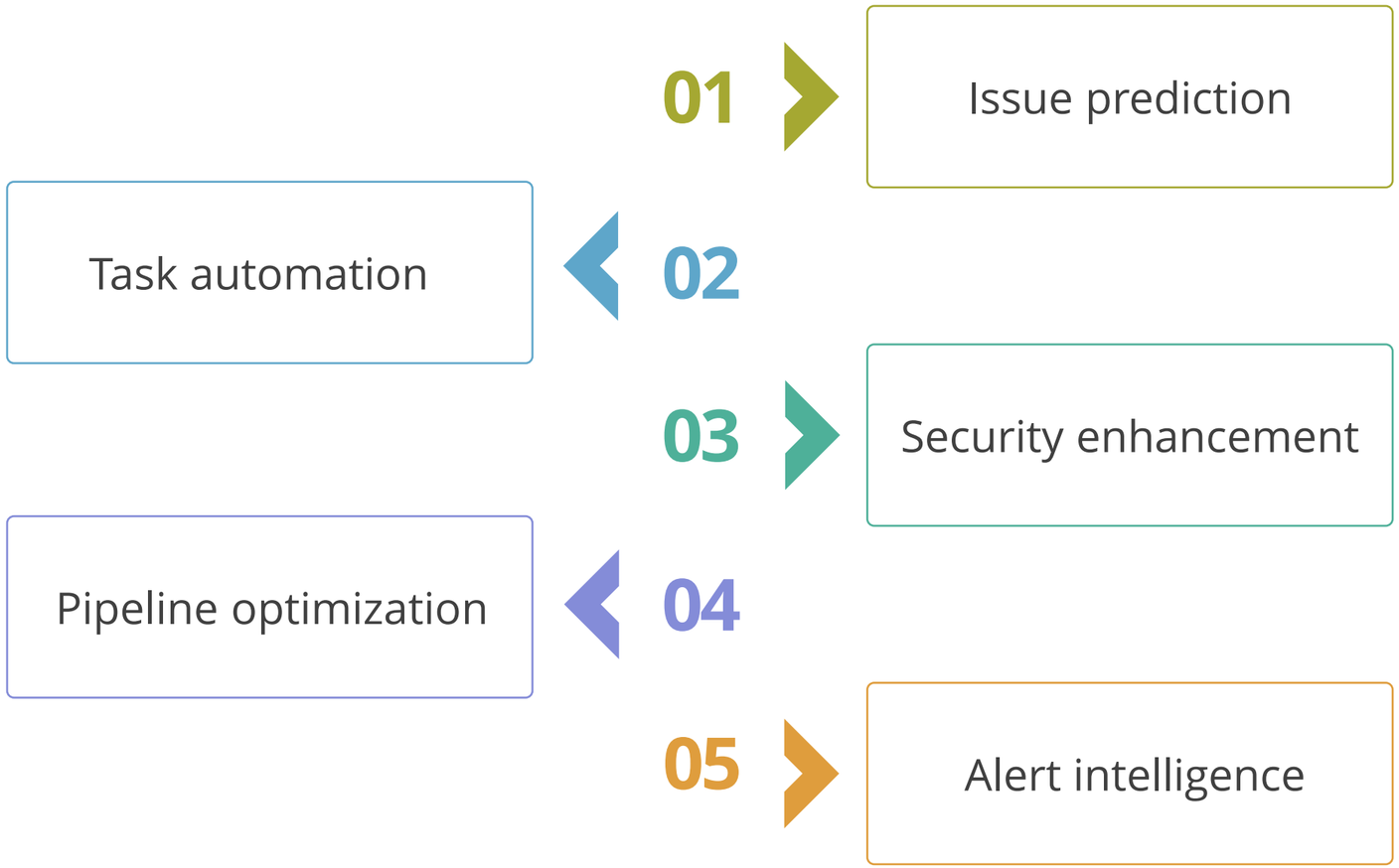
C. Serverless computing

D. Traditional IT ticket-based provisioning

Overview of AI and ML Integration in DevOps

How AI and ML Integrate with DevOps?

AI and ML enhance DevOps by introducing intelligent automation into the software development lifecycle. They analyze vast amounts of data from code commits, test results, logs, and system performance to enable:



Why AI and ML Are Essential for DevOps?

AI and ML are essential for DevOps because they bring intelligence, speed, and automation to the increasingly complex software delivery process. They help teams:

Accelerate innovation through data-driven insights

Improve reliability through predictive analytics and smart monitoring



Detect and resolve issues faster than manual methods

Enhance security with real-time anomaly detection

Automate repetitive tasks across the CI/CD pipeline

Key Use Cases of AI in DevOps

Integrating AI into DevOps enables intelligent workflows, driving smarter, faster, and more adaptive software delivery. Below are the key use cases where AI enhances DevOps processes:

- Test automation
- CI/CD automation
- Code recommendation
- Process improvement
- Reviewer recommendation
- Monitoring enhancement
- Anomaly detection
- Root cause analysis
- Vulnerability analysis

Benefits of Using AI in DevOps

AI empowers DevOps by driving smarter, faster, and more secure software delivery through automation and data-driven insights. The benefits include:



Quick Check



A DevOps team struggles with frequent false alerts and delayed issue identification during deployments. They want to enhance efficiency, reduce manual tasks, and resolve incidents faster using data-driven insights. Which DevOps enhancement best addresses this situation?

A. Implementing version control for infrastructure scripts

B. Adopting GitOps for deployment precision

C. Migrating CI/CD tools to a new platform

D. Using AI/ML for alert intelligence and anomaly detection

Exploring Internet of Things(IoT)

What Is Internet of Things (IoT)?

It is a system of interrelated things, including computing devices, mechanical and digital machines, objects, animals, or people equipped with unique identifiers.



It enables data transfer over a network without requiring human-to-human or human-to-computer interaction.

Components Used in IoT

A robust IoT system relies on foundational hardware components that enable sensing, processing, and communication. Two such essential components are:



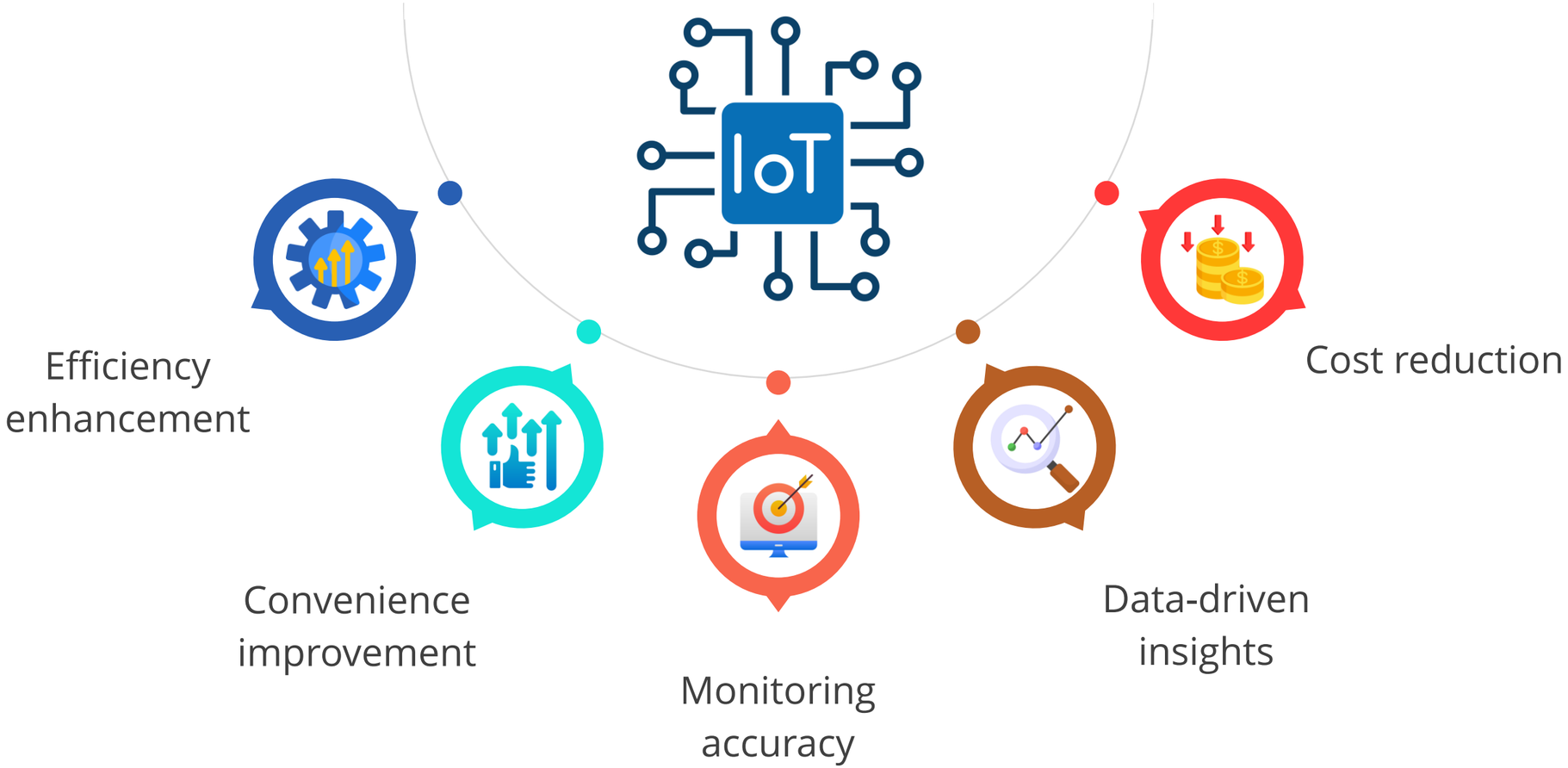
Low-power embedded systems: Correctly describes their role in energy-efficient computing.



Sensors: Accurately described as tools for collecting environmental data.

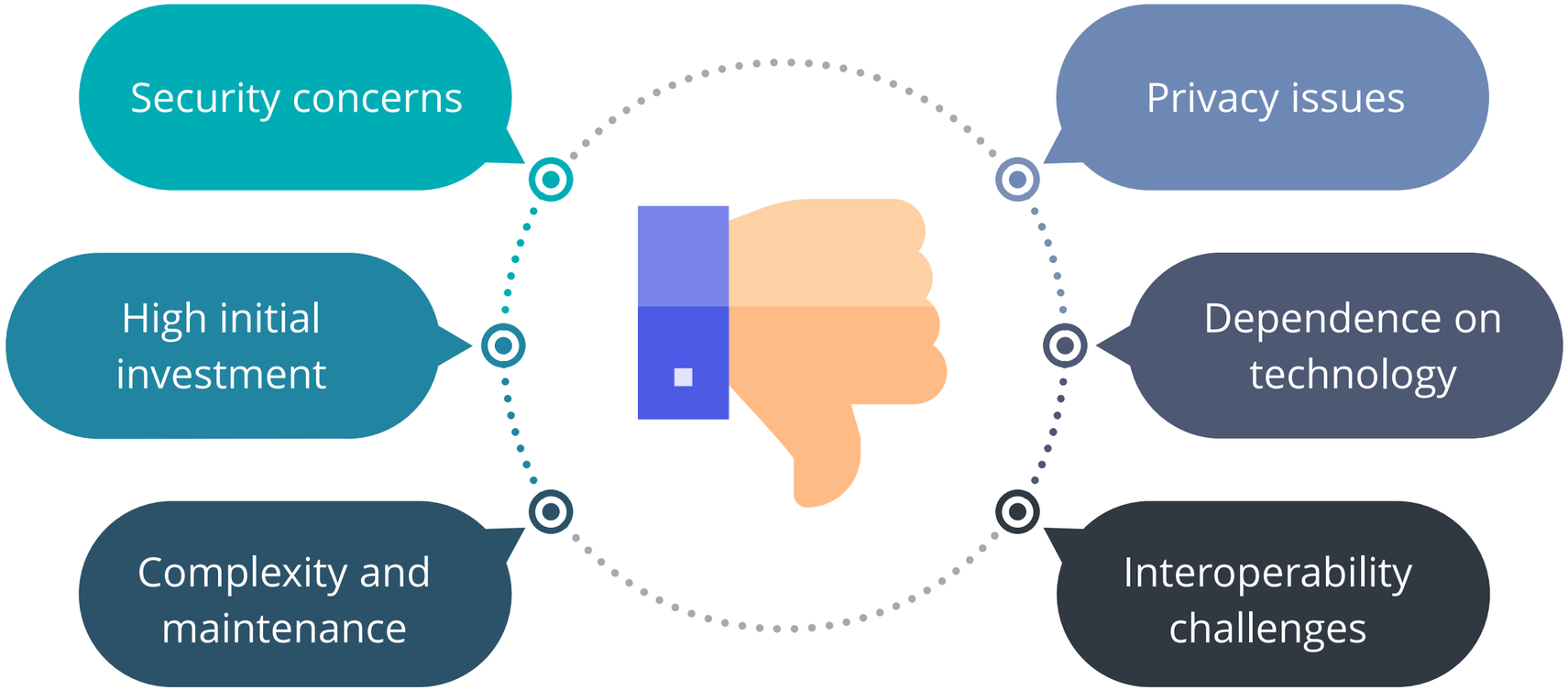
Advantages of IoT

IoT brings transformative benefits by streamlining operations, enhancing decision-making, and improving overall quality of life. Below are the major advantages that help increase productivity:



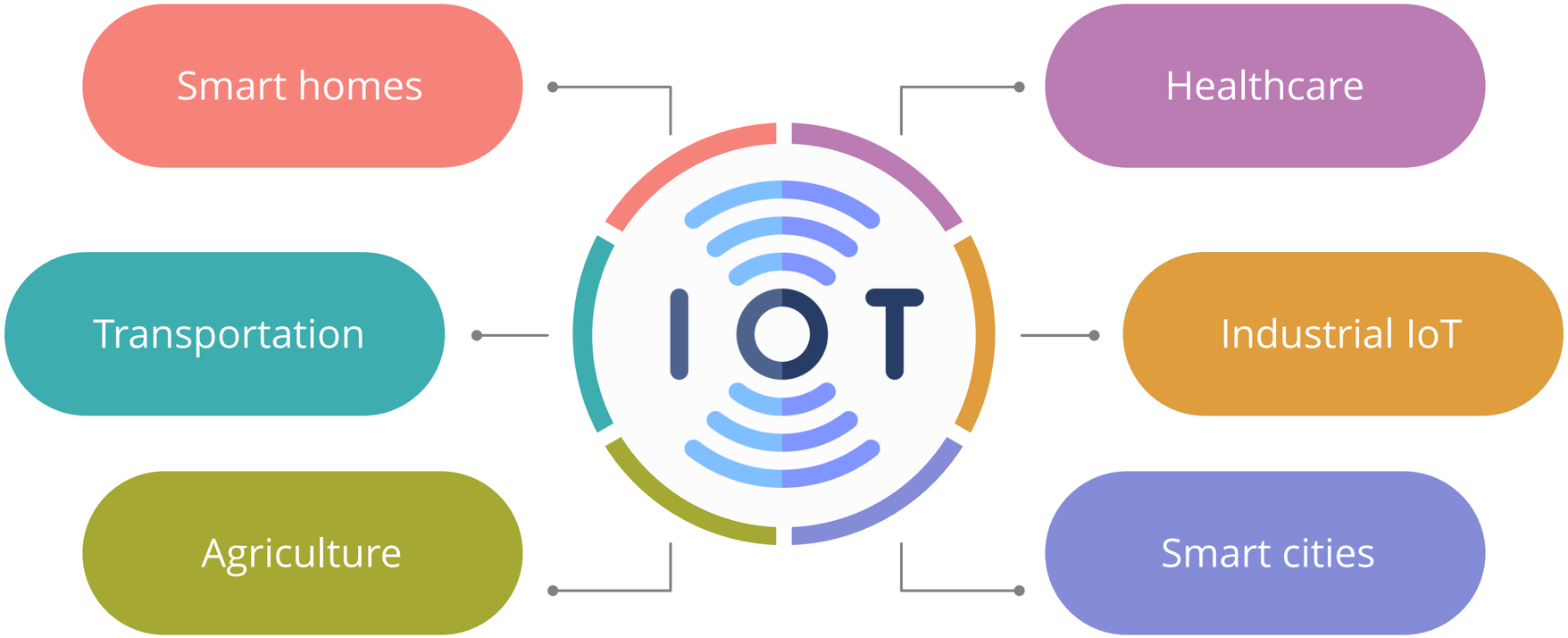
Disadvantages of IoT

IoT presents challenges that must be addressed to ensure secure, reliable, and sustainable implementation. The following are some of the most significant challenges:



Real-World Applications of IoT Across Industries

IoT technology is transforming diverse industries by enabling smarter environments, improving efficiency, and enhancing decision-making in real time. It plays a significant role in the following industries:



Future Trends in IoT

As IoT continues to evolve, emerging technologies and innovations are shaping its future, driving smarter, faster, and more secure connected systems. Below are some notable trends:

- Edge computing integration
- Energy-efficient design
- AI and ML convergence
- IoT security advancements
- 5G connectivity expansion
- Interoperability improvements
- Blockchain adoption

Quick Check



A smart factory monitors machine performance in real time and sends automated alerts to engineers without manual input. Which IoT feature is primarily responsible for enabling this functionality?

A. Automated sensing and processing

B. Data visualization tools

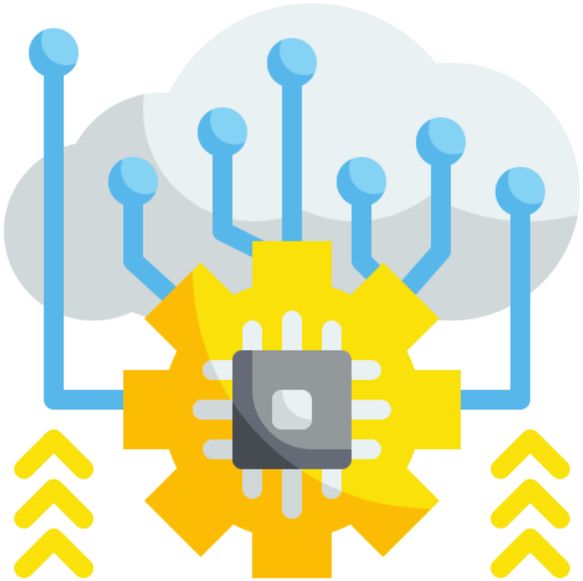
C. Cloud storage integration

D. Manual status reporting

Introduction to AIOps

What Is AIOps?

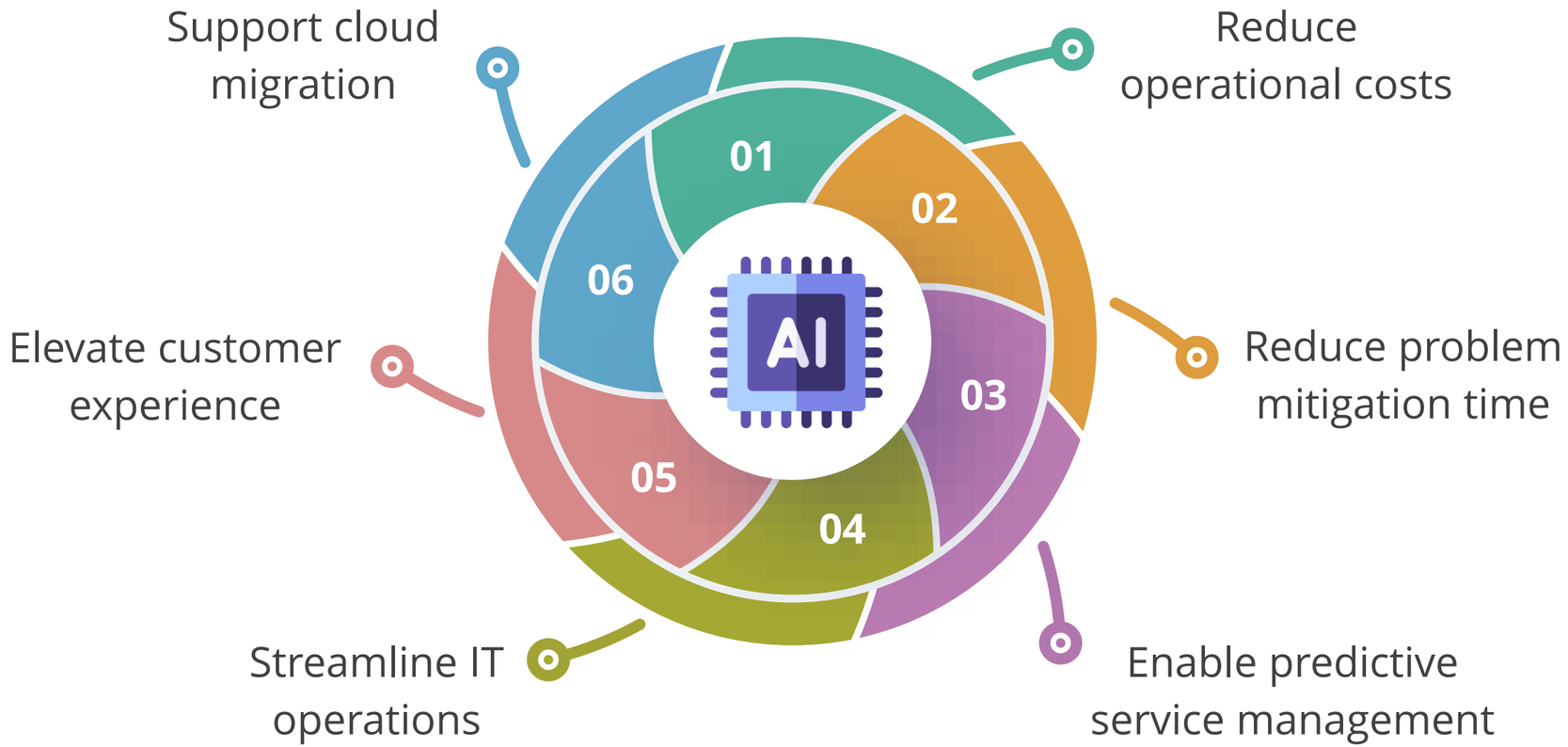
Artificial Intelligence for IT Operations (AIOps) is a cutting-edge approach that uses AI to enhance IT operations.



It combines machine learning (ML), natural language processing (NLP), and other AI technologies to automate and optimize various IT tasks, ensuring smoother and more efficient operations.

Why Is AIOps Important?

Organizations modernize IT operations, leverage AIOps to analyze and act on growing data volumes and drive measurable business value. Here are the key benefits:



How Does AIOps Work?

It uses machine learning and big data to streamline IT operations, enhance visibility, and automate responses across three key phases, as shown below:

Observe

Involves real-time data collection and analysis

Engage

Uses alerts and dashboards for team coordination

Act

Applies analytics and automation to maintain infrastructure health

Types of AIOps

AIOps offers organizations new ways to optimize operations and lower costs. It includes two distinct types of solutions, each designed to address specific operational needs, as outlined below:

Domain-centric AIOps

It focuses on a specific area, such as networking, applications, or cloud performance, and helps operational teams monitor and manage within that defined scope.

Domain-agnostic AIOps

This works across multiple domains, enabling IT teams to scale predictive analytics and automation by correlating data from diverse sources to generate broader business insights.

AIOps Use Cases

AIOps brings together ML, big data, and analytics to accelerate digital transformation across IT operations. Below are some notable use cases:

Application performance monitoring:

AIOps tools monitor complex, cloud-based applications by collecting and analyzing metrics across services and components.



Root cause analysis:

It identifies the underlying causes of incidents by correlating data from multiple sources in real time.



AIOps Use Cases

Anomaly detection:

It detects abnormal patterns in data to highlight potential system issues before they escalate.



Cloud automation and optimization:

It enables dynamic resource provisioning and cost-efficient scaling across cloud environments.



Application development support:

It improves software quality by automating code reviews and detecting bugs early in the development cycle.



Quick Check



A financial services company experiences frequent slowdowns in its online transaction platform during peak hours. To improve service reliability and customer experience, the IT team decides to implement a solution that can identify the root causes of performance issues in real time and trigger automatic responses. Which AIOps capability best addresses this requirement?

A. Task scheduling and job automation

B. Root cause analysis and anomaly detection

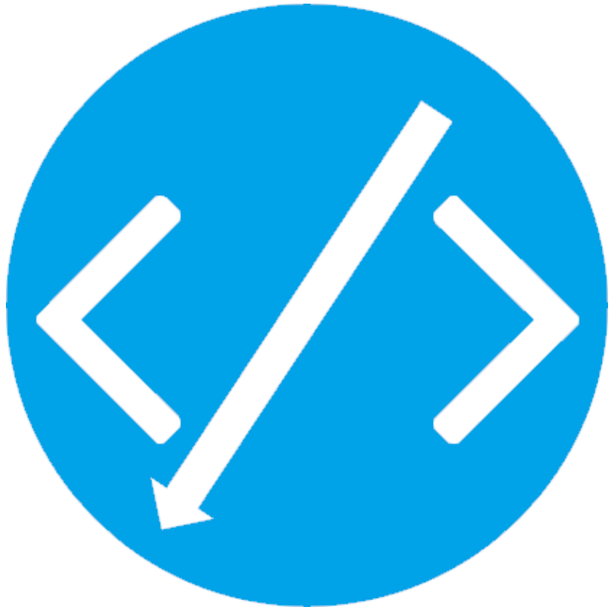
C. Hardware resource provisioning

D. Static threshold alerting

Overview of Low-Code Applications

What Are Low-Code Applications?

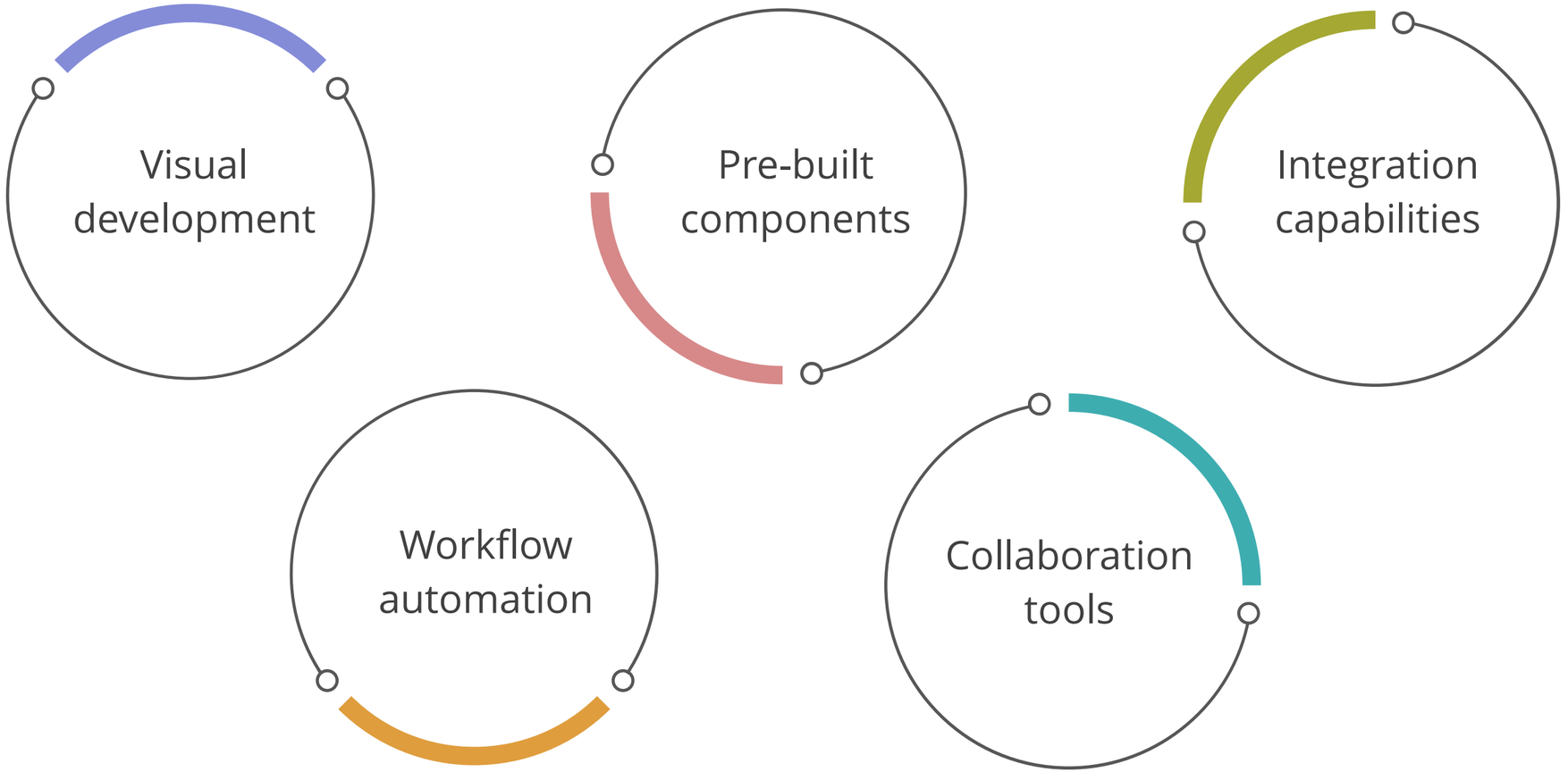
Low-code applications are built using visual development tools that require minimal hand-coding, enabling faster application delivery.



They allow both developers and non-developers to design workflows, integrate systems, and deploy apps with drag-and-drop interfaces and reusable components.

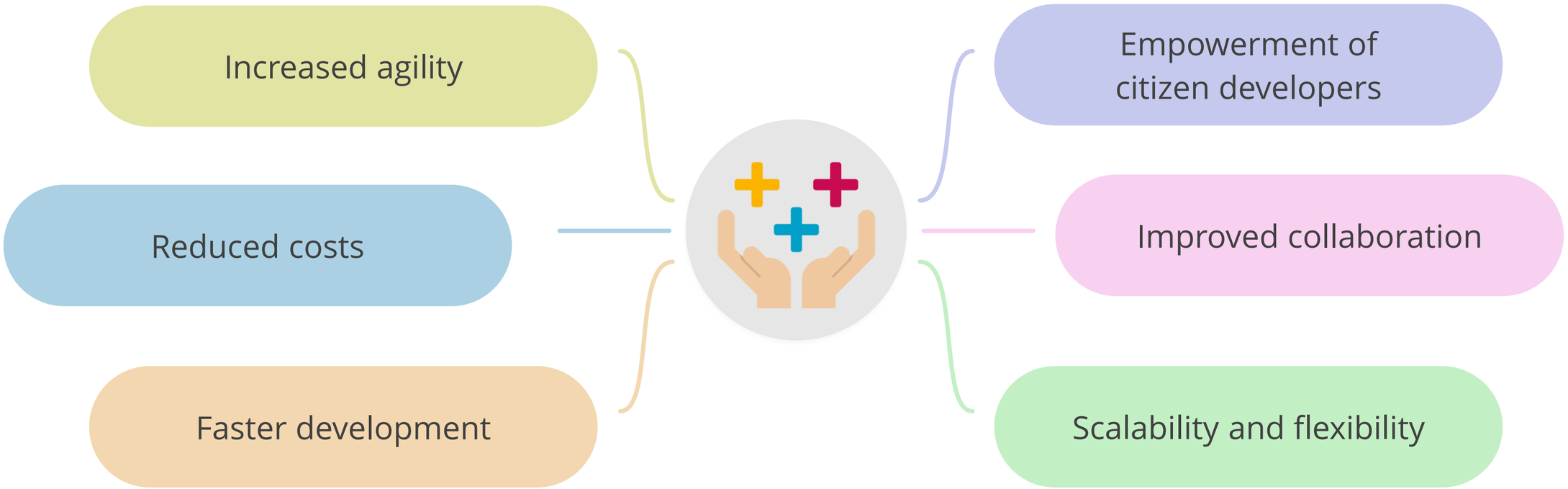
Core Capabilities of Low-Code Platforms

These are built on platforms that simplify software creation through intuitive design tools and rapid deployment features. Key capabilities that make these platforms fast, flexible, and collaborative include:



Benefits of Low-Code Applications

Low-code applications offer significant advantages to both developers and businesses by simplifying the software development process. The benefits include:



Use Cases of Low-Code Applications

Low-code applications support a wide range of use cases by enabling rapid development and automation of business-specific solutions with minimal coding effort. Below are some notable use cases:

Business process automation:

Automating repetitive tasks and workflows to improve efficiency and reduce manual errors



Customer Relationship Management (CRM):

Building custom CRM solutions tailored to specific business requirements



Use Cases of Low-Code Applications

Data management:

Creating applications for data collection, analysis, and reporting



Employee onboarding:

Developing onboarding portals and tools to streamline the hiring process



Field service management:

Designing applications to manage field service operations, including scheduling, dispatching, and tracking



Quick Check



A company wants to build a custom CRM solution to handle customer queries, automate follow-ups, and generate reports without hiring a large team of software developers. They also want to integrate it with their existing ERP system. Which type of platform would be most suitable for their needs?

A. Traditional high-code development platform

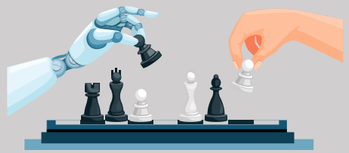
B. Low-code platform with integration and workflow automation features

C. CMS (Content Management System)

D. Standalone spreadsheet tools

Case Studies on Cloud and DevOps Implementations

Use Case 1: Netflix Infrastructure Scaling



Challenge



Solution



Outcome

As Netflix’s global user base expanded rapidly, the company faced the challenge of scaling its infrastructure to ensure seamless video streaming, high availability, and performance. It needed to manage vast amounts of data and support millions of concurrent users across diverse geographies, requiring a shift from traditional data centers to a more flexible, scalable architecture.

Source: https://netflixtechblog.com

Use Case 1: Netflix Infrastructure Scaling



Challenge



Solution



Outcome

- **Cloud-native strategy:** Netflix embraced a cloud-native model to boost scalability and resilience.
- **DevOps practices:** It adopted DevOps to streamline collaboration and accelerate delivery.
- **AWS migration:** It **s**hifted infrastructure to AWS for global scalability and reliability.
- **CI/CD automation:** It Implemented CI/CD pipelines to automate and speed up software releases.

Source: https://netflixtechblog.com

Use Case 1: Netflix Infrastructure Scaling



Challenge



Solution

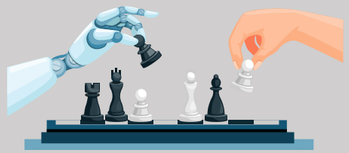


Outcome

- **Accelerated delivery:** Achieved faster and more reliable deployment cycles through CI/CD automation
- **Improved scalability:** Seamlessly scaled infrastructure to meet global demand using AWS
- **Enhanced experience:** Delivered consistent and uninterrupted content, improving user satisfaction

Source: https://netflixtechblog.com

Use Case 2: Amazon Deployment Optimization



Challenge



Solution



Outcome

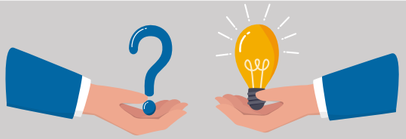
As Amazon’s e-commerce platform grew, the company faced pressure to reduce deployment time for new features and updates. Meeting customer expectations required a faster and more agile release process across its massive infrastructure.

Source: https://aws.amazon.com/builders-library/automating-safe-hands-off-deployments/

Use Case 2: Amazon Deployment Optimization



Challenge



Solution



Outcome

- **DevOps adoption:** Embraced CI/CD practices to accelerate software delivery
- **IaC tools:** Leveraged Terraform and Ansible for automated infrastructure provisioning
- **Cloud migration:** Transitioned to a scalable cloud environment for better flexibility and speed

Source: https://aws.amazon.com/builders-library/automating-safe-hands-off-deployments/

Use Case 2: Amazon Deployment Optimization



Challenge



Solution

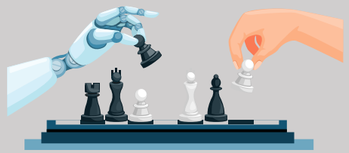


Outcome

- **Faster releases:** Reduced release cycles from weeks to minutes
- **Frequent deployments:** Enabled multiple reliable updates daily
- **High availability:** Maintained consistent system uptime to meet customer expectations

Source: https://aws.amazon.com/builders-library/automating-safe-hands-off-deployments/

Use Case 3: Airbnb Cloud Migration



Challenge



Solution



Outcome

Airbnb needed to scale its platform to handle increasing user traffic while optimizing resource utilization. The company sought a solution to enhance platform performance and meet global demand.

Source: https://aws.amazon.com/solutions/case-studies/innovators/airbnb/

Use Case 3: Airbnb Cloud Migration



Challenge



Solution



Outcome

- **Cloud migration:** Transitioned infrastructure to AWS to boost scalability and availability
- **DevOps integration:** Leveraged DevOps practices for automated infrastructure management and streamlined deployments

Source: https://aws.amazon.com/solutions/case-studies/innovators/airbnb/

Use Case 3: Airbnb Cloud Migration



Challenge



Solution

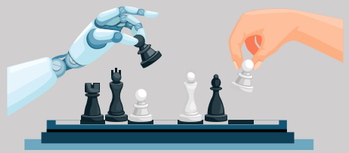


Outcome

- **Scalable operations:** Enabled dynamic scaling to handle fluctuating demand
- **Resource efficiency:** Improved usage of computing resources to reduce overhead
- **Operational gains:** Enhanced deployment speed and infrastructure responsiveness

Source: https://aws.amazon.com/solutions/case-studies/innovators/airbnb/

Use Case 4: Capital One Cloud Security Modernization



Challenge



Solution



Outcome

Capital One aimed to enhance IT operations and strengthen security posture while migrating its infrastructure to the cloud. The goal was to ensure scalability, reduce risks, and integrate security early into the development lifecycle.

Source: https://aws.amazon.com/solutions/case-studies/innovators/capital-one/

Use Case 4: Capital One Cloud Security Modernization



Challenge



Solution



Outcome

- **Cloud-native strategy:** Capital One adopted AWS cloud infrastructure to improve flexibility and scalability
- **DevSecOps integration:** Security was embedded into the CI/CD pipeline to identify vulnerabilities early.
- **Security tools:** It leveraged platforms like SonarQube and Snyk for continuous security monitoring and code analysis.

Source: https://aws.amazon.com/solutions/case-studies/innovators/capital-one/

Use Case 4: Capital One Cloud Security Modernization



Challenge



Solution

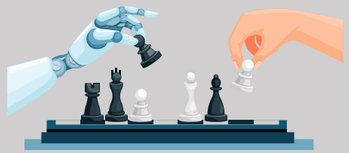


Outcome

- **Improved security posture:** Achieved proactive threat detection and mitigation
- **Operational efficiency:** Minimized risks and accelerated secure deployments
- **Seamless cloud transition:** Successfully migrated operations while maintaining compliance and performance

Source: https://aws.amazon.com/solutions/case-studies/innovators/capital-one/

Use Case 5: LinkedIn Performance and Reliability Optimization



Challenge



Solution



Outcome

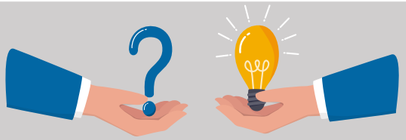
As LinkedIn’s user base and data volume grew rapidly, the platform faced increasing pressure to ensure high availability and performance. Managing large-scale, real-time data while minimizing latency and outages became a critical challenge. The team needed a robust solution to monitor infrastructure health and quickly resolve incidents. Providing an uninterrupted user experience was essential for global reliability and service continuity.

Source: https://www.linkedin.com/blog/engineering

Use Case 5: LinkedIn Performance and Reliability Optimization



Challenge



Solution



Outcome

- **DevOps implementation:** LinkedIn adopted DevOps practices for faster incident resolution and system resilience.
- **Monitoring stack:** It deployed tools like Prometheus, ELK Stack, and Grafana for real-time monitoring, logging, and visualization.
- **Incident management:** It built proactive alerting systems to detect anomalies early.

Source: https://www.linkedin.com/blog/engineering

Use Case 5: LinkedIn Performance and Reliability Optimization



Challenge



Solution



Outcome

- **Improved visibility:** Achieved real-time system observability and performance tracking
- **Faster incident response:** Identified and resolved issues before they affected users
- **Reliable user experience:** Delivered smooth, uninterrupted service at scale

Source: https://www.linkedin.com/blog/engineering

Assisted Practice



Automating RDS Instance Launch Using Terraform and GitOps **Duration: 15 Min.**

Problem statement:
You have been assigned a task to automate the provisioning of an Amazon RDS instance using Terraform and GitOps. This task involves creating Terraform configuration files to define the RDS instance and building a GitHub Actions pipeline that deploys the configuration automatically on push.

Outcome:
By the end of this demo, you will be able to define and deploy an Amazon RDS instance using Terraform and GitHub Actions, enabling automated infrastructure provisioning through a GitOps workflow.

Note: Refer to the demo document for detailed steps:
01_Automating_RDS_Instance_Launch_Using_Terraform_and_GitOps

ASSISTED PRACTICE

Assisted Practice: Guidelines



Steps to be followed:

1. Set up Terraform configuration to define the Amazon RDS instance
2. Build the GitOps pipeline using GitHub Actions for deployment automation
3. Commit and test the pipeline to validate successful provisioning of the RDS instance

Key Takeaways

- DevOps is evolving to include new practices focused on speed, collaboration, scalability, and automation across the entire software delivery lifecycle.
- Platform engineering helps abstract infrastructure complexity by offering reusable tools, services, and self-service environments to development teams.
- GitOps enhances deployment workflows by treating Git as the single source of truth, enabling secure, auditable, and automated software delivery.
- Infrastructure as Code (IaC) transforms infrastructure provisioning by allowing teams to manage resources using version-controlled code, improving consistency and scalability.



Key Takeaways

- AIOps introduces intelligent automation using machine learning and analytics to improve monitoring, detect anomalies, and support faster incident resolution.
- Low-code applications empower both technical and non-technical users to rapidly develop business solutions with minimal coding, improving agility and reducing development overhead.
- IoT integration in DevOps brings new opportunities and challenges, requiring careful consideration of sensing, processing, security, and interoperability.



Thank You